

References

- [1] C. Botelho, A. Abad, T. Schultz, and I. Trancoso, "Speech as a biomarker for disease detection," *IEEE Access*, vol. 12, pp. 184487–184508, 2024.
- [2] D. Ghia, G. Ciravegna, A. Koudounas, M. Fantini, E. Crosetti, G. Succo, and T. Cerquitelli, "A concept-based approach to voice disorder detection," *arXiv preprint arXiv:2507.17799*, 2025.
- [3] L. Ilias and D. Askounis, "Explainable identification of dementia from transcripts using transformer networks," *IEEE J. Biomed. Health Inform.*, vol. 26, no. 8, pp. 4153–4164, Aug. 2022.
- [4] E. C. L. Leschly, O. Roesler, M. Neumann, J. Liscombe, A. Hosamath, L. Arbatti, L. H. Clemmensen, M. Ganz, and V. Ramanarayanan, "An exploration of interpretable deep learning models for the assessment of mild cognitive impairment," in *Proc. Interspeech 2025*, pp. 271–275, 2025.
- [5] S.-I. Ng, L. Xu, I. Siegert, N. Cummins, N. R. Benway, J. Liss, and V. Berisha, "An end-to-end overview of clinical speech AI," *IEEE/ACM Trans. Audio, Speech, Language Process.*, vol. 34, pp. 1016–1048, 2026.
- [6] N. Ntampakis, K. Diamantaras, I. Chouvarda, M. Tsolaki, P. Sarigiannidis, and V. Argyriou, "NeuroXVocal: Detection and explanation of Alzheimer's disease through non-invasive analysis of picture-prompted speech," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2025*, pp. 410–419, Springer, Sep. 2025.
- [7] L. Xu, J. Liss, and V. Berisha, "Dysarthria detection based on a deep learning model with a clinically-interpretable layer," *JASA Express Lett.*, vol. 3, no. 1, 2023.